RESEARCH METHODS FOR COMPUTING AND TECHNOLOGY (CT098-3-2-RMCT)

INDIVIDUAL ASSIGNMENT

PART 1 -PROPOSAL

Name	**Low Yi Bo**
TP Number	**TP076096**
Lecturer	**Dr. Hemalatha A/P Ramalingam**
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Limitations of Traditional ERP Demand Forecasting in Retail SMEs: An Interpretable TKAN-Based Approach

Low Yi Bo (TP076096)

TP076096@mail.apu.edu.my

Abstract – Malaysian retail Small and Medium Enterprises (SMEs) offer great contributions to the national economy; but they are subjected to notable operational volatility due to the limitation of traditional Enterprise Resource Planning (ERP) forecasting modules. In today's dynamic e-commerce environment, the traditional linear statistical models do not work well in this context of customer behavior, which is typically non-linear, leading to recurring forecast errors, hence being costly the "Bullwhip Effect." This research targets the technology gap by presenting a Temporal Kolmogorov-Arnold Network (TKAN) architecture with advanced interpretable approaches tested in other application domains with high volatility. The proposed system deploys a spline-based procedure whereby learnable activation functions are deployed on edges of the network in place of nodes to capture complex non-linear dependencies while providing a mathematical transparency to a system which was lacking in traditional black-box systems. By adding this layer of transparent intelligence to existing ERP environments, the solution hopes to reduce the rates of forecast error significantly, lower inventory holding costs, and generate the level of interpretable precision needed by SMEs to keep them competitive in a rapidly changing digital economy.

Key Terms

Demand Forecasting, Temporal Kolmogorov-Arnold Network (TKAN), Interpretable AI, Inventory Management, Enterprise Resource Planning (ERP), Bullwhip Effect, Non-linear Demand Patterns

1. Introduction

SMEs play an important operational part in the supply chain and inventory management industries, particularly in the Malaysian retail and e-commerce industries. The importance of SMEs In Malaysia, SMEs have been identified as the backbone of the Malaysian economy, and they account for 38.4% of GDP in 2024 (DOSM, 2025). The efficiency of the national supply chain will play a critical role in supporting these companies since logistics and inventory management are the key operational pillars of the supply chains. Specifically, good inventory management is an important cost cutting lever that impacts cash flow and overall financial health of these businesses. In a case where margins are narrowing, the ability to reduce the cost of holding whilst increasing turns has become a significant success factor. Rahman and Rahim (2024) continue to explain that as these industries keep expanding their effectiveness in maintaining economic impact has been nothing measure than its production capacity but through the application of supply chain technology. As a result, economic prosperity for the industry now relies on the precision of their inventory planning skills.

However, the consistency of the efforts of planning inventory in the Malaysia retail SMEs are being threatened by the volatility of the market. Several factors affecting the resilience of supply chain in Malaysian E-Commerce note systematic challenges in a dynamic operating environment (Aun et al., 2025). In such a scenario, the demand patterns may be unstructured and non-linear, that would be difficult to model using the conventional forecasting tools. Conventional statistical forecasting tools, such as moving averages, exponential smoothing and ARIMA are designed to work with the linear trends and stationary patterns, but not effective in modeling complex, volatile and non-linear

demand behaviors, which may lead to repetitive mistakes in forecasting and inventory matching (Chowdhury et al., 2025). To mitigate such complexities, it has become a trend for Malaysian SME to seek digital supply chain solution to expand their businesses (Rahman & Rahim, 2024). Recent empirical research shows that ERP systems have been deployed by Malaysian SMEs as a part of these digitalisation initiatives (Hong et al., 2023). Grobler-Dębska et al. (2022) note that traditional ERP forecasting instruments are grounded in fixed static statistical models that are not adequate to deal with dynamic market volatility.

In the face of these emerging problems, the forecasting errors are increasingly seen as systemic risks, not just estimating errors. Previous studies of supply chains have shown that even small forecast deviation problems that are felt at the retail end of the supply chain will tend to grow larger in terms of chaos in orders and inventorying supply chain wide; this is called the Bullwhip Effect (Lee et al., 1997). Forecasting errors in the linear statistical model will continue to accumulate under volatile or non-stationary demand conditions instead of correcting them, ultimately leading to absurd accumulation of inventory, outages of stock, and general disruption of the supply chain. More recent researches on forecasting have shown how the error magnification in this sort of forecasting usually a result of the existence of the non-linear nature of the demand by linear models, the residuals from linear forecasting models are constantly amassing and hence the accuracy of planning is affected (Chowdhury et al., 2025). Therefore, in previous literatures, the accuracy of forecasts has been indicated as the problems of adaptive modeling instead of static trend estimation.

With the growing dependence on modern day forecasting applications, a need has been felt for advanced methods of Machine Learning to cope with the non-linear aspects of customer demand. While deep learning models such as Long Short-Term Memory (LSTM) networks have been developed to capture these long-term dependencies, they are a 'black box' with no interpretability, which is the fundamental drawback in the trust of SME managers and stakeholders preventing their practical adoption. To solve this, recent computational mathematical developments have given rise to the Temporal Kolmogorov-Arnold Network (TKAN). Unlike standard neural networks, TKANs replace linear fixed weights with learnable B-spline activation functions on the edges of the network as they thus allow for capturing highly volatile, non-stationary temporal relations while generating mathematically transparent rationales for their predictions [26, 27, 28, 29]. Despite the capability of such architectures to reduce error magnification during forecasting, while providing user trust, these methodologies have not been able to be implemented in a methodical fashion to overcome the structural limitations of legacy ERP-based inventory planning systems.

Therefore, this research is aimed at addressing the gap between linear statistical limits and precision forecasting by proposing the use of an interpretable TKAN architecture module to replace legacy ERP forecasting modules. The proposed approach uses the distinctive structure of the TKAN to enable processing of histories of sales and volatile external market factors simultaneously in such a manner that they preserve long-term memory while not forfeiting the possibility of interpretability symbolically. By providing a mix of accurate non-linear demand forecasts and the transparent mathematical formulas, the proposed model ensures that decision-makers can trust the automated inventory recommendations instead of having to manually override the model in case of doubt. In the end, this approach aims to address the Bullwhip Effect problem, address the failure in fulfillment by the findings by Kamarudin et al. (2025) and achieve sustainable profitability for the Malaysian SMEs.

2. Problem Statement

Malaysian Retail SMEs that depend on existing ERP systems are increasingly facing high levels of errors when forecasting which fundamentally impacts the efficiency of operations and the market effect. This critical problem is currently the prime concern in Malaysia whereby SMEs are encouraged to adopt digital supply chain technology in order to survive and build their business, but often times, fails to attain the stability needed in operation to survive and develop their business (Rahman & Rahim, 2024). This problem brings a severe and widespread effect on the target users, this is a huge unbalance in inventory, which directly affects the business margin. These errors are specifically responsible for causing the "Bullwhip Effect", that either results in costly inventory that freezes up capital or stockouts that result in lost revenue and customer dissatisfaction. The underlying reason for this mass failure is that the existing available solutions are architecturally built for stable, predictable environments, and have absolutely no computational complexity to handle the volatile, non-linear demand patterns that are the nature of the modern e-commerce market.

From a technical point of view, the inefficiencies in forecasting are a result of the specific analytical techniques driven within many legacy ERP demand planning modules. Prior ERP implementation studies shows that such systems often use univariate statistical analysis tools, such as ARIMA and Exponential Smoothing which are used to forecast future demand basing mainly on past linear patterns (Grobler-Dębska et al., 2022). Although these methods worked fine under stable market conditions, recent research in forecasting confirms the fact that linear models do not work well mathematically to model non-linear, sparse, or highly volatile demand patterns (Chowdhury et al., 2025). Consequently, residual demands fluctuations are systematically, instead of being actual meaningful signals, treated as random noises. While standard deep learning models can capture these complex fluctuations, models are considered "black box" like and business managers therefore cannot trust those automated outputs. Therefore, this research motivates the solutions of both technical limitations and interpretation by proposing a Temporal Kolmogorov-Arnold Network (TKAN) architecture. This approach breaks the legacy modules for forecasting and replaced them with an interpretable, mathematically transparent engine that can be used to process volatile e-Commerce data with high predictive accuracy within existing ERP-based inventory planning systems.

3. Research Aim

The aim of this research is to propose an interpretable TKAN model adapting computational mathematics to resolve legacy retail ERP limitations.

4. Research Objectives

1. To examine the impact of recurring forecast errors on Malaysian Retail SMEs using current ERP systems.
2. To analyze the technical limitations of traditional statistical models in handling non-linear demand patterns.
3. To propose a new integrated system that incorporates an interpretable Temporal Kolmogorov-Arnold Network (TKAN) within legacy ERP frameworks.

5. Research Questions

1. What are the impacts of recurring forecast errors on Malaysian Retail SMEs using current ERP systems?
2. What are the technical limitations of legacy models when dealing with non-linear demand patterns?
3. How can the proposed Temporal Kolmogorov-Arnold Network (TKAN) architecture be integrated into legacy ERP systems?

6. Research Significance

To optimize inventory management cost efficiency for Malaysian SMEs – The significance of this research is to deliver a tangible solution that halts the financial losses incurred as a result of the "Bullwhip Effect." Recent research by Chua et al (2025) highlights that logistics optimization and inventory costing is a vital survival mechanism for Malaysian SMEs who operate on thin margins. By dramatically improving the accuracy of forecasts, with this research, companies can minimize the high cost of unsold inventory and avoid lost revenue due to stockouts. This has a direct positive impact on their financial performance, which is in effect unlocking cash flow which is currently locked away in poor planning for reinvestment and sustainable growth.

- **To bridge the gap between linear statistical theory and non-linear market reality** – The significance of this research is to prove that advanced computational mathematics can be successful in situations where conventional ERP systems have been unsuccessful. Examples of the Malaysian retail industry For many companies, manual estimates or simple history, which does not have the ability to adjust to real-time market dynamics, are still in use (Shamima et al., 2025). This research creates vital real-world proof that by using the Temporal Kolmogorov-Arnold Network (TKAN) architecture, there is a much more robust way to model the complicated, volatile models of today's, complex e-commerce data as opposed to the legacy statistical methods that are used in them right now.

- **To empower decision-makers with data-driven precision instead of intuition** – The importance of this research is to replace intuitive guesswork with interpretable mathematical proof. Research on Malaysian SMEs, managers still continue to use "gut feeling" rather than data analytics because they do not have access to advanced AI tools and, often times, do not trust "black box" AI models (Domracheva, 2025). This framework transforms banal history into concrete and actionable intelligence. Because the TKAN has symbolic interpretability, it outputs exactly the mathematical rationale for its forecasts. This helps to ensure that SME managers are not blindly trusting an algorithm to help them make sense of their data and build the crucial trust needed to prevent manual overrides, and reduces the risk of human error, making the business nimble and resilient (Jalil et al., 2024).

7. Overview of the Proposed System

7.1 Existing System

The functional workflow of the existing system is characterized by the fact that it is dependent on conventional ERP forecasting modules. As described by Grobler-Debbrowska et al. (2022), conventional models in these systems are univariate statistical logic which uses such algorithms as the ARIMA algorithm or the Exponential Smoothing algorithm. The conventional workflow showed in Figure 1 begins with the processing of the historical sales data as processing with the assumptions of linearity and stationarity. Under these conventional settings, the ERP generates a demand forecast purely based on historical trends which then directly finds its use for inventory planning with no validation.

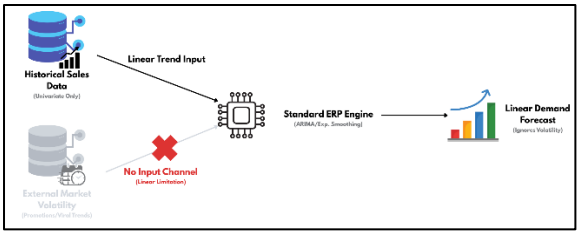

Figure 1 Existing ERP-Based Demand Forecasting Process

7.2 Limitation of Existing System

Despite the automation provided by these systems, they have a "Linearity Limitation." According to Chowdhury & et al. (2025), conventional statistical models are built with a stable environment and cannot perform on data that is non-linear, sparse, or volatile. Consequently, the current system contains a logical "Blind Spot" and is unable to process exogenous variables such as market shocks and viral trends. This inability to model non-linear trends causes an error in the forecast, which leads to constant underestimating of sudden demand peaks by the system and thus directly to the "Bullwhip Effect" as described by Lee et al. (1997).

Moreover, Figure 2 shows the Architectural Limitation found by Grobler-Dębska et al. (2022). Conventional ERPs tend to be "Closed Systems" where demand forecasting module is directly embedded within the software architecture. This renders the system unable to interface with external high performance computing tools. As the forecasting logic is directly hard-coded within the legacy ERP architecture, it is not possible to directly update it with modern deep learning functionalities forcing SMEs to stick to conventional statistical models despite the availability of better technology.

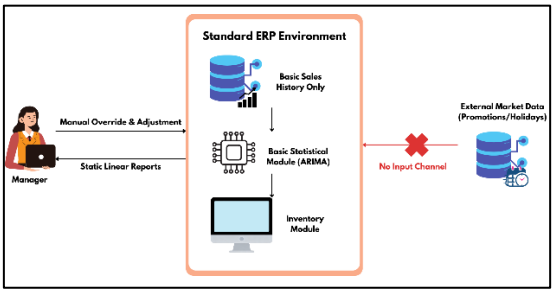

Figure 2 Architectural Limitations of Traditional ERP Forecasting Systems

7.3 Proposed System

To overcome the limitations of linear models and 'black-box' nature of standard Neural Networks, this research presentations proposed Temporal Kolmogorov - Arnold Network (TKAN) Enginer. As shown in the revised process flow, the TKAN does not differentiate between linear and non-linear patterns. Instead, it processes the historical sales and outside market factors at the same time. By placing learnable B-spline activation functions on the network edges instead of on fixed nodes, the TKAN handles long-term temporal memory in conjunction with continuous mapping of exact mathematical relationships of variables. The end result is a very precise forecast of demand coupled with a transparent symbolic formula so that the managers are able to interpret and trust the non-linear corrections in the market.

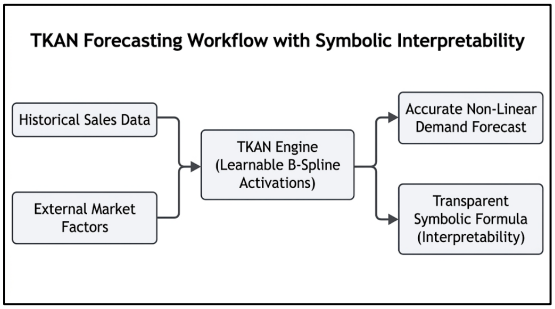

Figure 3 TKAN Forecasting Workflow with Symbolic Interpretability

7.4 System Architecture

In order to overcome the rigidity of architecture, proposed system is implemented as separate 'Intelligence Layer'. Incorrectly, the conventional monolithic architecture allows the forecasting module to be integrated into the main server, whereas the Decoupled Architecture put forth in this proposed research work deceives. In this configuration, the TKAN Forecasting Engine is operating independently of the main ERP system. This way, since the advanced computational tasks of the TKAN do not impact the running of the main ERP server that processes normal business transactions.

The integration is realized through a Multivariate Data Feed where the ERP is the data provider. As illustrated in the system design, the ERP performs the task

of sending historical sales data with external event indicators to the Intelligence Layer. The TKAN engine then can processes the feed in parallel and return both an Optimized Demand Forecast and Interpretable Symbolic Output back to the Inventory Module of the ERP. The system design thus bypasses the limitations of the "Closed System" as listed by Grobler-Débrska et al. (2022) in order to make possible the implementation of the mathematical robustification of the forecasting system adapted from recent TKAN literature (Genet and Inzirillo, 2024; Liang and al, 2025) putting into practice in the current legacy IT infrastructure of the organization.

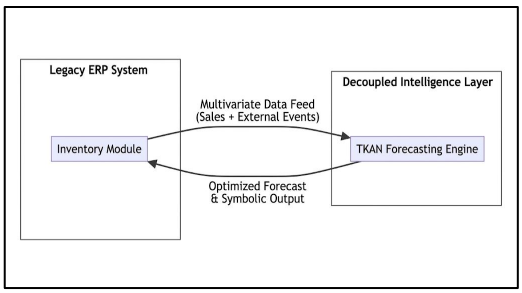

Figure 4 Decoupled TKAN Architecture Integrated with ERP

8. Conclusion

The work that has been done has been the exploration about the challenges of demand forecasting in legacy ERP system adopted by Malaysian retail SMEs which is ineffectiveness of the conventional statistical models in the volatile and non-linear demand pattern. Through extensive literature review of recent literatures and practices it was identified that the traditional demand forecasting approach by using ARIMA is not effective in current retail and e-commerce environment. In order to overcome these problems, an interpretable Temporal Kolmogorov-Arnold Network (TKAN) Demand Forecasting model was proposed in this research. Departing from traditional black-box deep learning, the TKAN relies on learnable B-spline activation functions to unify the processing of historical and external data to deliver a systematic solution to improving the accuracy of demand forecasts while also providing the mathematical transparency needed to create the user trust in the ERP supported inventory planning process.

However, the proposed framework remains on a conceptual level and has not been empirically validated yet, using real-world retail data. The proposed TKAN architecture is very much dependent on the availability and quality of historical demand data and data maturity of SMEs about the record keeping of promotional activities and external demand factors. Future work should focus on implementing the proposed system with real-world e-commerce data, designing suitable data preprocessing workflows, and empirically testing the feasibility of the proposed system using traditional ARIMA-LSTM baselines and using already-established error measures. This will serve as validation of the practicability of spline-based architectures as well as their superiority in real-world ERP applications.

(2647 words)

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Appendix

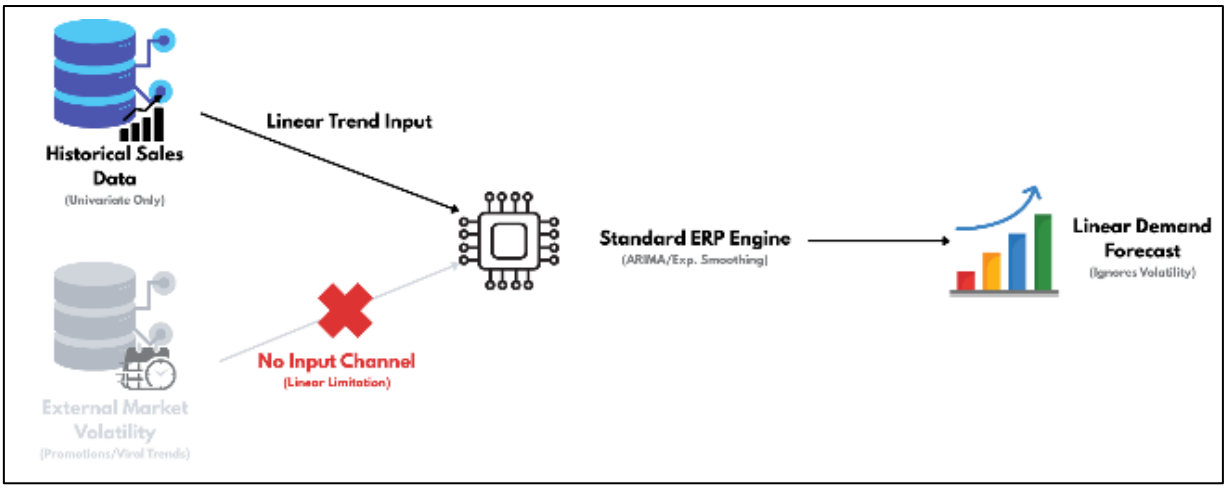

Figure 5 Existing ERP-Based Demand Forecasting Process

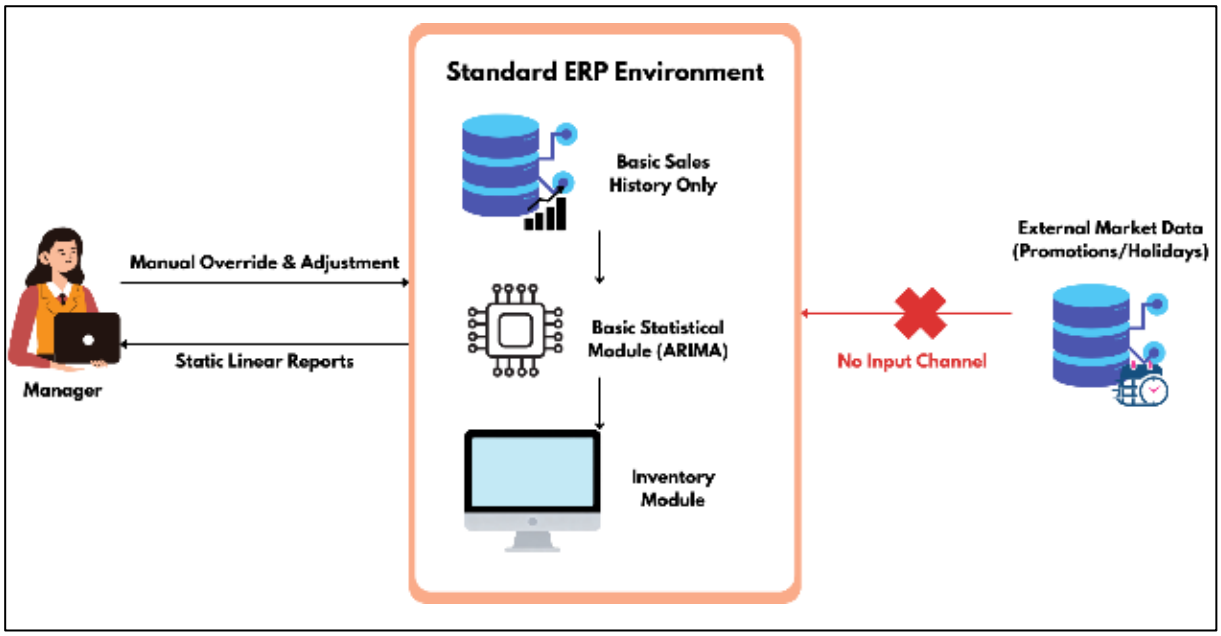

Figure 6 Architectural Limitations of Traditional ERP Forecasting Systems

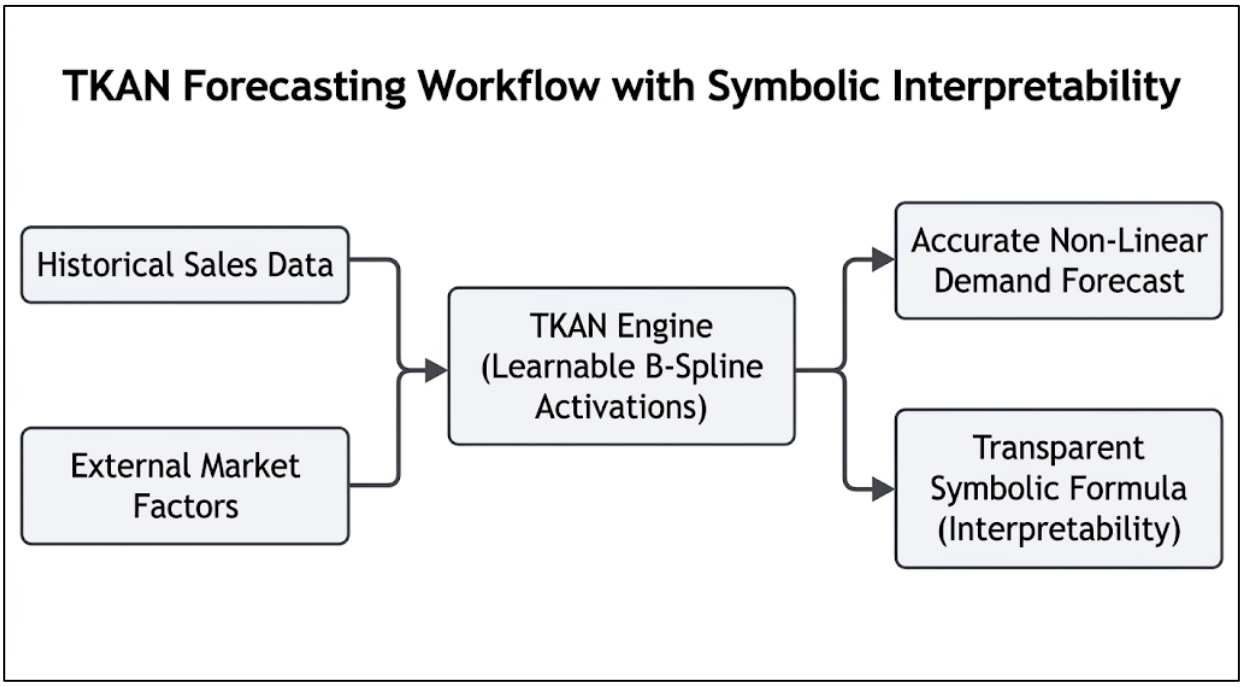

Figure 7 TKAN Forecasting Workflow with Symbolic Interpretability

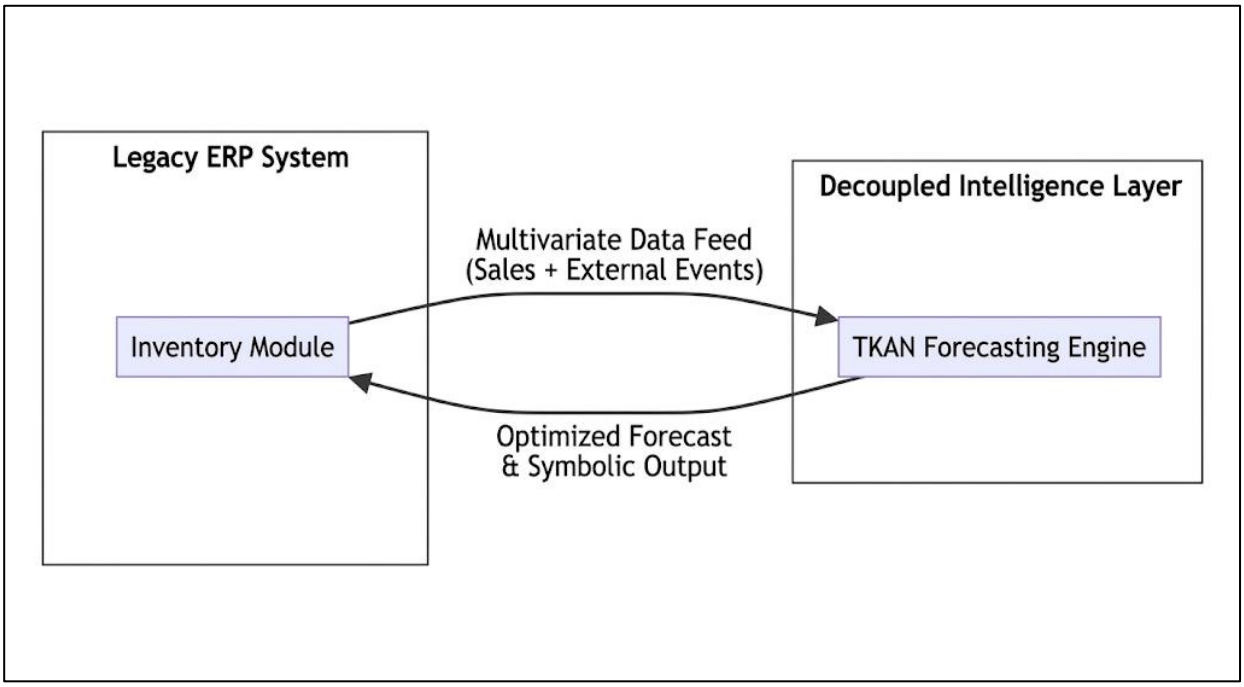

Figure 8 Decoupled TKAN Architecture Integrated with ERP